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RESEARCH-ARTICLE

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Abstract

Bearings play a crucial role in rotating machinery, and their failure significantly impacts performance and safety. Early fault detection is challenging due to weak signals, especially under variable operating conditions. Deep learning techniques, such as Convolutional Neural Networks (CNN) and attention mechanisms, have shown great potential in improving fault detection accuracy in recent years. However, handling complex fault patterns and extracting early fault features remain challenging. To address these issues, this paper proposes a new method that combines Temporal Convolutional Networks (TCN), CNN, and Multi-Head Attention. TCN captures short-term dependencies, CNN extracts long-term features, and Multi-Head Attention enhances focus on key fault-related features. The model was tested on the Case Western Reserve University (CWRU) bearing fault dataset and compared with baseline models, showing a significant improvement in accuracy, achieving 99.3%. This study provides valuable insights for handling complex and multi-fault scenarios, offering essential implications for industrial applications.

CCS Concepts

• **Networks** → Network algorithms; Network economics.

Keywords

fault detection, multi-head attention, temporal convolutional networks, convolutional neural networks

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1 Introduction

Bearing fault diagnosis plays a crucial role in industrial equipment. As a key component in rotating machinery, a bearing's operational condition directly affects the system's overall performance, safety, and service life [1]. Bearing failures can lead to equipment downtime, production interruptions, and even safety incidents, resulting in significant economic losses and potential casualties [2]. Therefore, timely and accurate bearing fault diagnosis ensures stable equipment operation and reduces maintenance costs. With the advancement of artificial intelligence, researchers have proposed various signal processing, machine learning, and deep learning-based methods to improve the accuracy and efficiency of fault diagnosis. However, existing bearing fault diagnosis methods still face several limitations, primarily their poor sensitivity to weak early-stage fault signals and inadequate feature extraction capabilities [3]. Conventional fault diagnosis methods rely on manually extracted features, heavily influenced by expert knowledge and subjectivity, limiting their ability to comprehensively and accurately assess early bearing failures. Single-feature extraction methods often lack generalizability, making capturing failure characteristics across different locations, types, and severity levels in complex systems challenging. Although modern signal processing techniques, such as wavelet transform and sparse representation, have somewhat enhanced fault signal extraction, they suffer from parameter sensitivity and high computational complexity [4].

The contributions of this paper are summarized as follows:



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(1) Integration of Temporal Convolutional Networks (TCN) and Convolutional Neural Networks (CNN) for Enhanced Feature Extraction. The proposed model leverages TCN for efficient time-series data processing while incorporating CNN's long-term feature extraction capabilities, improving fault feature extraction performance.

(2) Multi-Head Attention for Adaptive Feature Selection. The multi-head attention mechanism dynamically selects fault-related critical features, enhancing the model's focus on early-stage fault characteristics while effectively suppressing redundant information.

(3) A Comprehensive Diagnostic Framework. This study introduces an integrated framework that combines temporal feature learning with attention mechanisms. It aims to unify early fault warning and identification and provides theoretical support for real-time monitoring and fault prevention.

2 Related Work

Traditional bearing fault diagnosis methods primarily rely on signal processing techniques, such as vibration signal analysis, acoustic recognition, and temperature monitoring [5] [6]. These methods involve deploying sensors to collect vibration signals from bearings, followed by spectrum analysis or Short-Time Fourier Transform (STFT) to identify fault characteristic frequencies and types. These approaches are suitable for detecting bearing faults under various operating conditions, providing online and offline monitoring capabilities, and enabling accurate early-stage fault diagnosis [7] [8].

With the rise of deep learning, its applications in bearing fault diagnosis have gained increasing attention. Deep learning techniques automatically learn high-level feature representations from raw data, overcoming the limitations of handcrafted feature design inherent in traditional methods. For instance, CNNs have become widely used to diagnose bearing faults. By automatically extracting local features from vibration signals, CNNs significantly enhance fault diagnosis accuracy and robustness [9] [10]. The integration of channel and spatial attention mechanisms has further optimized CNN learning, allowing the model to extract fault features more effectively. Attention mechanisms have also been employed to assign importance to different input data, maintaining high diagnostic accuracy even under complex operating conditions [11]. These studies showcase considerable progress in applying deep learning and attention mechanisms to bearing fault diagnosis, offering promising avenues for more efficient and robust diagnostic methods. This paper proposes an early fault diagnosis method that integrates TCN, CNN, and multi-head attention mechanisms to address these issues. This approach mitigates existing research limitations in early fault feature extraction, multi-fault pattern recognition, and cross-dataset generalization, providing an efficient solution for fault prediction and diagnosis in complex failure scenarios.

3 Methodology

This section outlines the methodology of the proposed fault diagnosis model, which integrates (TCN), (CNN), and a Multi-Head Attention mechanism. The TCN branch captures long-range temporal dependencies in time-series data, while the CNN branch focuses

on extracting spatial features. The Multi-Head Attention mechanism enhances the model's ability to dynamically focus on essential aspects of the data, allowing for more flexible feature learning. The TCN and CNN branches are fused to combine both temporal and spatial features, which are then processed by the attention mechanism. The resulting features undergo average pooling and are passed through fully connected layers to generate predictions. The model is trained using a cross-entropy loss function and the Adam optimizer. This fusion of techniques improves the model's handling of complex, multiscale fault scenarios, making it highly effective for fault diagnosis in time-series data.

3.1 TCN

This study adopts a one-dimensional, fully convolutional network (TCN). It utilizes causal convolution to ensure that the network only relies on present and past time steps, preventing information leakage from future data points. Additionally, the dilated convolution module enables the expansion of the receptive field without increasing the number of parameters, thereby enhancing the model's ability to capture long-term dependencies efficiently.

3.2 CNN

A CNN model is highly effective for processing structured grid data, such as time-series signals or images. It excels in feature extraction and pattern recognition by capturing spatial hierarchies of features. The core components of a CNN include convolutional layers, activation functions like ReLU, pooling layers such as average or max pooling, and fully connected layers that transform extracted features into final predictions. This paper applies dimensionality reduction to the input feature maps by computing the average value within each region. This approach decreases data dimensionality and computational complexity while preserving essential features, thereby improving the network's efficiency and generalization capability.

3.3 Multi-head attention

The multi-head attention mechanism learns input representations from different subspaces in parallel, helping the model capture more semantic information and adaptively allocate weights to focus on essential features, thereby improving its feature selection capability. When combined with TCN, the model becomes more flexible and efficient in handling long-term dependencies. The proposed multi-head attention comprises multiple one-head attention mechanisms.

3.4 TCN-CNN multi-head attention fusion

The TCN-CNN Multi-Head Attention combines the advantages of TCN and CNN by integrating the multi-head attention mechanism. This fusion enhances the model's ability to model time-series data, particularly excelling in handling long time-series and multi-scale features. The TCN-CNN Multi-Head Attention combines the advantages of TCN and CNN by integrating the multi-head attention mechanism. This fusion enhances the model's ability to model time-series data, particularly excelling in handling long time-series and multi-scale features, as shown in Figure 1.

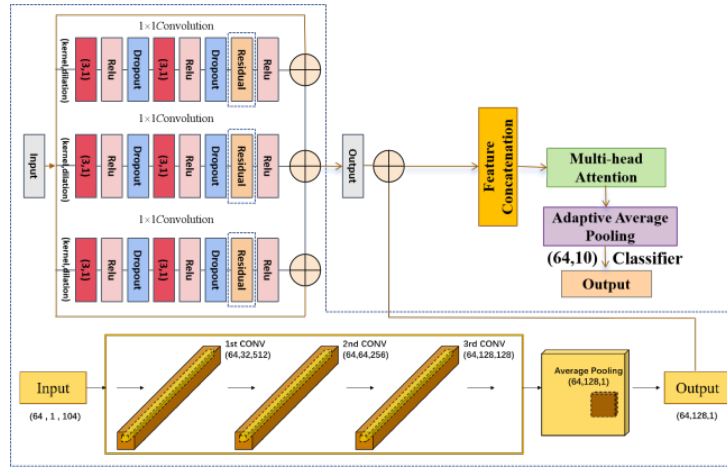


Figure 1: TCN-CNN multi-head attention model fusion.

3.5 Implementation pipeline

3.5.1 Input stage. The input to the model is a tensor X , representing the raw time-series data. The dimensions of X are $N \times C \times T$, where N is the batch size, C is the number of input channels and T is the length of the time sequence. The raw data is processed to ensure that it is suitable for the deep learning model, and the input tensor carries the information required for fault diagnosis.

3.5.2 TCN branch. The TCN branch is essential for learning the temporal dependencies present in the input data. The TCN uses dilated convolutions, allowing the network to capture long-range dependencies without exponentially increasing the parameters. This is especially beneficial for time-series data, where patterns may span over a long period. Each temporal block in the TCN branch performs convolution operations to process the input sequence, with the output sequence length denoted by T' .

$$T' = \text{ConvPool}(X) \quad (1)$$

3.5.3 CNN branch. The CNN branch is designed to capture spatial features within the time-series data. The convolution operation applied to the input sequence generates intermediate feature maps highlighting essential patterns. These features are beneficial for identifying local anomalies or faults. The CNN branch performs a standard convolution operation, producing an intermediate feature F_{cnn} :

$$F_{cnn} = \text{Conv}(X) \quad (2)$$

3.5.4 Feature fusion. After the TCN and CNN branches independently process the data, their outputs are concatenated along the channel dimension to form a combined feature set. This fusion of temporal and spatial features allows the model to leverage both types of information simultaneously, ensuring a more comprehensive representation of the input data.

$$F_{\text{fusion}} = \text{Concatenate}(\text{TCN_output}, \text{CNN_output}) \quad (3)$$

3.5.5 Multi-head attention mechanism. The fused features are processed using the multi-head attention mechanism, a crucial step in the model. The multi-head attention mechanism allows the model

to focus on different parts of the input sequence at once, each head learning to attend to different subspaces of the input data. This ability to process the input in parallel helps capture a wide range of relationships and interactions that might be missed. The mechanism computes the queries Q , keys K , and values V using linear transformations of the fused features.

$$Q = W_q F_{\text{fusion}}, \quad (4)$$

$$K = W_k F_{\text{fusion}},$$

$$V = W_v F_{\text{fusion}}.$$

4 Experiments

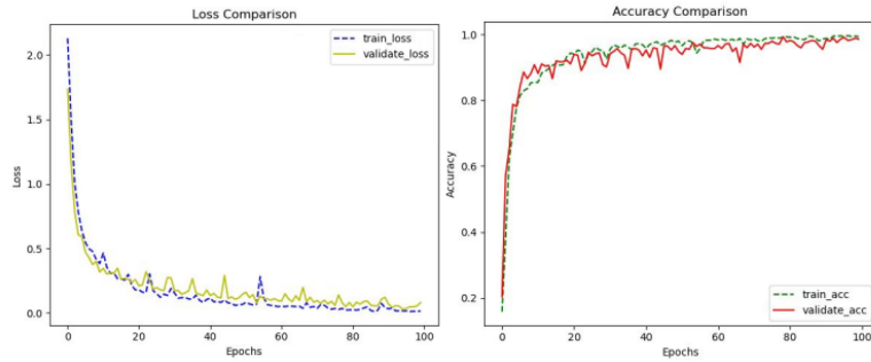
This section provides a detailed overview of the proposed fault diagnosis model's methodology, experiments, and results. The study utilizes the Case Western Reserve University (CWRU) bearing fault dataset, which is processed into time-series segments for model training. The model, which integrates TCN, CNN, and multi-head attention mechanisms, is trained and evaluated using Python and TensorFlow/Keras.

4.1 Dataset description and processing

The CWRU bearing fault dataset consists of 119,808 rows and 10 columns. Each row represents a time-series instance capturing the state or signal of a specific bearing. At the same time, each column corresponds to a distinct fault category, with labels designated for classification tasks. The dataset is partitioned into time-series segments, where each sample contains 1,024 time steps with an overlapping stride of 512. Consequently, each row includes 1,024 data points and an associated class label. The data is normalized to $[0,1]$ to optimize model training, ensuring consistency and improving convergence during learning. The generated samples are randomly shuffled after column-wise segmentation to mitigate bias from sequential dependencies. The dataset is then divided into a training set (70%), a validation set (20%), and a test set (10%). This partitioning strategy ensures that the model learns the distinctive features of various fault types, thereby enhancing classification performance. The model structure is shown in Table 1.

Table 1: Model architecture.

Layer	Description	Value
TCN Layer	Number of Stack	3
	Number of filters Dilation rate	[32, 64, 128]
	Kernel size	1, 2, 4
	Dropout rate	5
CNN Layer	Number of convolutional layers	3
	Conv/feature maps	(1, 32), (1, 64), (1, 128)
Multi-Head Attention Layer	Embedding dimension	4
Output Layer	Output feature dimension	10
Training	Batch size	32
	Learning Rate	0.0003

**Figure 2: Comparison of loss values and accuracy rates during the training and verification processes.**

4.2 Evaluation metrics

The outcomes of both setups are evaluated using accuracy as the primary performance metric. Accuracy indicates how the model predicts correctly across all classes by calculating the ratio of correct predictions to the total number of predictions. where TP, TN, FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively.

$$Accuracy(Acc) = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

5 Results and Discussion

This study proposes a fault diagnosis model that combines TCN, CNN, and multi-head attention mechanisms. Experimental results demonstrate that this model significantly outperforms other comparative models in terms of accuracy and stability, achieving an accuracy of 99.3%. Result analysis shows that the model performs excellently when handling complex time-series data, primarily due to the integration of TCN's capability in processing temporal data and CNN's efficient feature extraction ability. Furthermore, the multi-head attention mechanism enhances the model's feature learning capacity and allows a more flexible focus on important information within the input data.

5.1 Results analysis

Figure 2 show the changes in training loss (train loss), validation loss (validate loss), training accuracy (train accuracy), and validation accuracy (validate accuracy) across training epochs. The training and validation loss trends are generally consistent, with the validation loss fluctuating slightly, indicating no significant overfitting occurred during the training process.

5.2 Ablation studies

Table 2 presents the accuracy comparison of different models. The accuracy of the TCN-Attention model is 91.29%, which is slightly higher than the CNN-Attention model's 90.85%. The TCN-CNN model achieves an accuracy of 94.2%, demonstrating the benefits of combining TCN and CNN. The best-performing model is the TCN-CNN MultiHead Attention model, which achieves an accuracy of 99.3%. This indicates that the Multi-Head Attention mechanism significantly enhances model performance and plays a crucial role in feature extraction and representation learning.

6 Conclusion

This paper proposes a novel bearing fault diagnosis method that integrates TCN, CNN, and Multi-Head Attention mechanisms, offering a unique approach to processing time-series data efficiently. The technique significantly enhances the accuracy of early bearing

Table 2: Accuracy comparison of different models.

Model	Accuracy (%)
CNN-Attention	90.85
TCN-Attention	91.29
TCN-CNN	94.20
TCN-CNN	99.30
Multi-Head Attention	

fault diagnosis by utilizing TCN for efficient sequence modeling, CNN for long-term feature extraction, and Multi-Head Attention for improved focus on critical fault characteristics. Experimental results on the CWRU bearing fault dataset demonstrate outstanding performance, achieving an accuracy of 99.3%, particularly in handling complex fault patterns and multi-fault scenarios. This performance surpasses traditional and deep learning models, showcasing the model's superior feature extraction and representation learning capabilities. Moreover, the proposed model accurately detects early, subtle fault signals while minimizing redundant information, thereby improving generalization and stability. Despite these advantages, the model's high computational requirements, especially for large-scale datasets, may pose challenges such as extended training times and increased memory consumption. Future work will focus on optimizing the model to reduce computational overhead and further testing on additional real-world datasets to validate its scalability and robustness.

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